

Analysis and Research on UART Communication Protocol

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Abstract—This article is a comprehensive research paper on the Universal Asynchronous Receiver Transmitter (UART) communication protocol, aiming to familiarize relevant personnel with the principles and applications of the Universal Asynchronous Receiver Transmitter protocol and contribute to the further development of UART research. UART is the most fundamental communication protocol in digital communication. After continuous research and improvement by scientists such as Gordon Bell, it still has its unique value and is widely used in various fields such as the Internet of Things, industrial control, and healthcare. In the second chapter of this article, the basic principles section introduces the hardware structure of UART and the implementation method of UART communication protocol. The hardware structure of UART includes a UART transmitter, UART receiver, baud rate generator, and control circuit. The UART communication protocol relies on transmitting data frames, which include start bits, data bits, parity bits, and stop bits. At the same time, this article also analyzes the advantages and current challenges of UART. In the third chapter of this article, some typical research achievements based on UART by modern scientists are listed and their research value is analyzed. I believe that with the development of digital technology in the future, the UART protocol can be integrated into more microprocessors to obtain more efficient communication methods and achieve greater value.

Keywords—UART, communication protocol, application

I. INTRODUCTION

Universal Asynchronous Receiver Transmitter(UART) protocol is one of the most fundamental communication protocols in digital communication. This technology has undergone decades of transformation and is still widely used in various fields such as the Internet of Things, industrial control, and medical treatment. It is of great significance for the development of modern industry and technology. Therefore, it is necessary to systematically study the UART protocol. In Chapter 2, this article focuses on analyzing the various principles of UART, reviewing its development history, elaborating on the composition and functions of each hardware structure of UART, as well as the implementation of UART communication protocols. This passage examines the benefits and difficulties associated with UART. UART was initially created by Gordon Bell, an engineer at Digital Equipment Company (DEC), and his colleagues during the 1970s. By the end of the 1990s, UART had become a popular means of communication between personal computers and other devices [1]. The hardware structure of UART includes a UART transmitter, UART receiver, baud rate generator, and control circuit [2]. The UART communication protocol relies

on transmitting data frames, which include start bits, data bits, parity bits, and stop bits [3]. UART has the advantages of simplicity and ease of use, flexible and controllable speed, and saving pin resources, but it also faces challenges such as low transmission rate, high hardware requirements, low data accuracy, and difficulties in unconventional transmission [4, 5]. In Chapter 3, this article lists typical research achievements of the UART protocol in the fields of medical, telecommunications, and aviation in recent years. For example, Hu et al. extended the hardware circuit by expanding the Bluetooth module of the UART interface, achieving simultaneous access to various medical devices, and improving the efficiency of doctors in real-time obtaining patient health indicators [6]. The research in this article is beneficial for relevant personnel to become familiar with the principles and applications of UART, and contributes to the further development of UART research.

II. THE BASIC PRINCIPLES OF UART

A. Development History

The emergence of the prototype of UART began in the early 1970s. In the late 1960s and early 1970s, computer systems urgently needed a more flexible and efficient communication interface. At that time, the commonly used form of serial communication interface was asynchronous serial communication interface, but this interface had certain defects, such as slow transmission speed and unreliable data transmission.

Due to the limitations of existing serial communication interfaces, Gordon Bell and his colleagues from Digital Equipment Company embarked on research for a new interface. The objective was to attain dependable and high-speed data transmission while ensuring that the interface possessed excellent flexibility and compatibility [1].

In the process of developing UART, they adopted many innovative technologies. The most important technology among them was to use a baud rate generator to generate accurate clock signals to ensure the accuracy and reliability of data transmission. In addition, they had improved the data frame structure and error detection mechanism of the asynchronous serial communication interface to improve the reliability of data transmission.

After years of effort, Gordon Bell and others finally successfully developed UART, which became one of the most used serial communication interfaces in computer systems at that time. UART has not only been widely used in DEC

computer systems, but has also become one of the most basic communication interfaces in modern computer systems.

Based on Bell et al.'s research, National Semiconductor launched the first UART chip NS16550 to support high-speed communication, which became one of the most popular UART chips in the market at that time. This is a popular UART variant with a 256-byte FIFO buffer, which can significantly improve the efficiency of data transmission.

Towards the end of the 1990s, UART had established itself as a conventional computer interface and was extensively utilized for communication between personal computers and other devices. Despite being gradually supplanted by USB in certain domains, UART continues to hold significance in numerous embedded systems [7].

In recent times, with the advent of Industry 4.0 and the Internet of Things (IoT), the utilization of UART in IoT devices and sensor networks has been on the rise, and improvements in the development of UART continue to be made [8].

B. UART hardware structure

The hardware structure of UART includes a UART transmitter, UART receiver, baud rate generator, and control circuit [2]. The hardware structure diagram is shown in Fig.1.

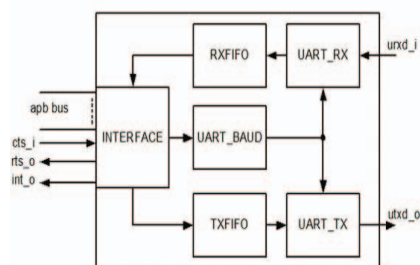


Fig.1. The hardware structure diagram [9]

1) UART transmitter

The UART transmitter is an important component of UART hardware. Its main function is to convert parallel data transmitted by the CPU to UART into serial data, and use special bits at the beginning and end of each data word to synchronize the transmitter and receiver.

The send buffer is used to store the data bytes to be sent. The function of the transmitter shift register is to convert parallel data into serial data for transmission to the communication line. When the CPU wants to send data to an external device, it writes the data to the send buffer. Then, the transmitter reads the data from the send buffer and loads it into the shift register. The shift register serializes data bit by bit and transfers it to the communication line. At the beginning and end of each data byte, the transmitter also inserts a start bit and a stop bit, so that the receiver can correctly parse the data. The transmitter can also use optional check bits to detect whether data has been transmitted correctly.

2) UART receiver

The main function of the UART receiver is to receive serial data from the communication line and convert it into parallel format for CPU processing.

When serial data is transmitted from the communication line, the receiver will load the data into the shift register. The shift register will deserialize the data bit by bit and store it in the receive buffer. At the beginning and end of each data byte, the transmitter also inserts a start bit and a stop bit, so that the receiver can correctly parse the data. The receiver can also use optional check bits to detect whether the data has been transmitted correctly. Once the data is stored in the receive buffer, the CPU can read and process it. If the receive buffer is full, the receiver will stop receiving serial data until the CPU reads and retrieves some data before continuing to receive it.

3) Baud rate generator

The baud rate generator is one of the necessary components for the normal operation of UART. The baud rate generator in UART is responsible for generating the clock signal required for each bit cycle to ensure communication between the sender and receiver at the same rate. The baud rate generator controls the rate of UART, enabling the sender and receiver to correctly interpret the transmitted data. If the baud rate does not match, the receiver will not be able to receive data correctly, resulting in communication errors. The higher the baud rate is, the greater the transmission speed is. The common baud rates used for UART include 9600bps, 115200bps, etc.

4) Control circuit

The control circuit controls the receiver, such as detecting whether the reception buffer is full, detecting whether there is data transmission on the communication line, etc.

C. Implementation of UART Communication Protocol

UART communication transfers data in packets called data frames, which contain four elements: a start bit, a data bit, a parity bit, and a stop bit [3]. The start bit marks the beginning of the data frame, while the stop bit signals its end. To ensure data accuracy, check bits may also be added. The UART communication protocol diagram is shown in Fig.2.

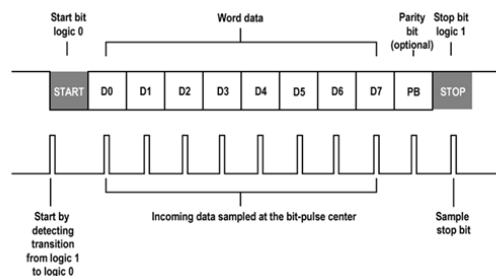


Fig.2. The UART communication protocol diagram [10]

1) Start Bit

During UART data transmission, the sender and receiver do not have a uniform clock. To avoid data transmission errors, a start data transmission flag bit needs to be added before data reception. UART sends data by T-ray and receives data by R-ray. In idle state, TX and RX bus are both high levels. Therefore, when the UART is idle, a low level indicates that it can start receiving data. The receiver detects the presence of a low-level signal after the stop bit or during idle time to determine whether data needs to be received.

2) Data Bits

The number of data bits in UART can be programmed according to the communication requirements, generally between 5-9 bits. Because UART transmits slowly, if meaningless data exists in the data bits, communication efficiency will be reduced. Therefore, in order to ensure the communication efficiency, it is necessary to select an appropriate number of data bits according to the characteristics of data transmission. Before data transmission, both sender and receiver need to unify the number of bits to avoid data miss or multiple connections.

3) Parity Bits

When using UART for data transfer, check bits can be added to the data frame to help verify whether there are errors in the data transfer received by the receiver. UART uses odd or even checking and places the check bit before the stop bit. UART usually uses RS-232 or RS-485 for long-distance transmission, but data transmission problems may occur due to interference. To solve this problem, we can add check bits so that the receiver can check the received data. If a check error indicates an error in the sending process, the data received is invalid. Otherwise, the data can be received and further transmitted. Before data transmission, both the sender and the receiver need to configure, including choosing whether to add check bits in the data frame.

4) Stop bits

In UART, there is a stop bit at the end of each data frame. By default, the stop bit is 1 bit high. However, the stop bit can also be configured as 1.5 or 2 bits depending on the usage requirements. If the stop bit does not appear at the end of the data frame as expected, a frame error will occur.

D. Advantages and Challenges of UART

1) Advantages of UART

The primary advantage of UART is its ease of use. The UART protocol is known for its ease of implementation and versatility, making it a popular choice for integration into various chips and systems [11]. Additionally, the speed of UART can be adjusted to suit the specific requirements of different applications [4]. It supports a variety of different baud rates and can be adjusted to achieve the best transmission speed in different applications. UART has a long transmission distance and can support a longer transmission distance than SPI and I2C because it uses differential signal transmission and has stronger anti-jamming capability. UART only requires two pins (one input and one output) compared to other serial communication interfaces, thus saving valuable IO resources. Because of these advantages, UART has been widely used: because UART is a standard serial protocol, it is widely used in the communication between various devices, including computers, embedded systems, smartphones, sensor networks, and so on.

2) Challenges faced by UART

The transmission distance of UART communication is limited by such factors as voltage and signal attenuation. External devices such as converters or amplifiers are needed to enhance the signal during long-distance transmission.

Low transmission rate: Compared with modern communication protocols, UART communication has a lower transmission rate [5]. Although the transmission rate can be increased by increasing the baud rate, it increases the complexity and cost of the system.

High hardware requirements: UART communication requires more hardware resources, including send and receive buffers, clocks, counters, etc. This may result in increased cost and complexity of the system.

Low data accuracy: UART communication is based on serial transmission, which has fewer data bits, so there are some limitations in the transmission accuracy.

Unconventional transmission difficulties: UART communication can only support simple point-to-point communication and requires reconfiguration of parameters. If unconventional communication, such as broadcasting or multicast, other communication technologies are required.

III. APPLICATION AND ANALYSIS

Because of its advantages of good reliability, high flexibility and simple structure, UART communication protocol has been widely used in many fields, and under the improvement of researchers, UART communication protocol has been developed in more ways of application. This paper will carry out detailed examples and analysis of the application of UART communication protocol in the following three fields of medical, telecommunications and aviation, and summarize the case literature of specific applications of UART in these fields.

A. The application of UART in the medical field

Due to the scalability of its interface, UART has been widely used in medical signal transmission, helping processors to quickly obtain a variety of different information.

In his article, Hu Bangmin applies UART to the general interface of medical equipment. He expands the hardware circuit by expanding the Bluetooth module of the UART interface. This allows him to realize that a variety of different medical devices can be accessed at the same time and the detection information on medical equipment can be obtained and sent to the medical information system. By doing so, he improves the efficiency of doctors obtaining patients' physical health indicators in real time [6].

In their article, Vladislav team applied UART to the interface development of EMG recorders, receiving detection data from different hardware at the same time through the UART register, and the UART interface was connected to a PC to exchange data through the internal boot record of the microcontroller to realize the real-time collection and processing of electrical signals generated by muscle cells [12].

B. The application of UART in the field of telecommunications

Due to its communication protocol transmission and reception advantages, UART has been widely used in telecommunications to improve circuit hardware performance, help hardware update and performance improvement.

Rizka's team applies UART to improve the performance of the Internet of Things, proposes a fast and scalable UART communication protocol by enhancing the scalability of the UART protocol, combines UART with FPGA to reduce the possibility that the communication performance may be destroyed when UART performs bit tapping technology, and realizes the function of improving the node performance, speed, and scalability of the IoT platform [13].

In their article, BinduSwetha's team applied the UART protocol to a test VLSI circuit. They utilized the BIT

technology based on the UART standard to replace the traditional testing method. By connecting the CPU and peripherals through the UART protocol, they designed an eight-bit UART module to facilitate short-distance communication between the CPU and peripherals. The UART's receiver and transmitter structure achieved the conversion of bytes and basic frames. Parity was added to perform simple error checking. Additionally, they employed BIST technology based on the UART standard to improve chip test efficiency and reduce test time [14].

Stepan Harutyunyan's team, in their article, applied UART to the design of a new interface design controller, which realized automatic detection of baud rate and automatic adjustment by enhancing the UART controller architecture with automatic parameter detection and introducing the instruction set, while fully adapting to the function of connected devices and working without manual configuration [15].

C. The application of UART in the field of aviation

Due to its advantages in full-duplex transceiver communication, UART has been widely used in space communication to help spacecraft collect and quickly process various data.

A. S. Keerthi Nayani's three-person team, in their article, applied the UART protocol design to a part of telemetry in the field of avionics. They implemented its function in missiles and implemented the function of converting serial data and parallel data between serial data by applying UART to establish full-duplex communication. This helped to realize the role of forming duplex communication between the remote data acquisition unit (RDAU) and the main data acquisition unit (FPGA) [16].

In his article, Xing Shibo applied UART in avionics performance detection and diagnosis. He designed a multi-bus avionics communication module through the UART bus controller module transceiver function and FPGA technology. By using the bus analog transceiver combined with signal conversion, he realized the acquisition and transmission function of multiple different signals [17].

Chen Ren and Wang Haiying's team proposed in their article to apply UART technology to the internal communication of spaceborne instruments. Through the bus topology structure of UART transceiver circuit and multi-transmit and multi-receive, combined with triple module redundancy, TMR technology, and digital filtering technology, they aimed to reduce particle interference in the circuit, improve the fault tolerance rate and reliability of the system, and ensure the normal operation of internal wave communication in the complex space electromagnetic environment [18].

IV. CONCLUSION

This paper mainly studies and analyzes UART, a digital communication protocol technology, which is still widely used in various digital systems after decades of transformation. The UART communication protocol has an adjustable communication rate and can communicate between different devices, and has become one of the most common communication methods in fields such as medical, telecommunications, and aviation. Looking back at the development history of UART, UART faces challenges such as low transmission rate, high hardware requirements, and low

data accuracy while developing its advantages, so in the typical examples listed in this article, the authors improve the communication efficiency of the UART protocol and reduce the hardware requirements by optimizing and improving the protocol. In the future, with the development of digital technology, the UART protocol can be integrated on more microprocessors, thereby improving communication efficiency, and obtaining more efficient communication methods.

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